Replication Assignment

Punishment and Deterrence: Evidence from Drunk Driving

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1. Paper Information

Citation:

Hansen, Benjamin. 2015. "Punishment and Deterrence: Evidence from Drunk Driving." *American Economic Review*, 105(4): 1581–1617. https://doi.org/10.1257/aer.20130189

Link:

https://www.aeaweb.org/articles?id=10.1257/aer.20130189

Table being replicated: Table 2

TABLE 2—REGRESSION DISCONTINUITY ESTIMATES FOR THE EFFECT OF EXCEEDING BAC THRESHOLDS ON PREDETERMINED CHARACTERISTICS

	Driver demographic characteristics			**Police ex ante information**		
Characteristics	**Male**	**White**	**Age**	**Accident**	**Prior**	**PBT**
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. DUI threshold						
DUI	0.007	0.002	−0.165	−0.004	0.039	−0.0007
	(0.005)	(0.005)	(0.167)	(0.004)	(0.071)	(0.0005)
Mean (at 0.079)	0.792	0.852	34.9	0.089	0.139	0.090
Controls	No	No	No	No	No	No
Observations	95,111	95,111	95,111	95,111	95,111	60,485
Panel B. Agg. DUI threshold						
AGG DUI	−0.001	0.003	0.049	0.003	0.008	0.0002
	(0.001)	(0.003)	(0.123)	(0.004)	(0.006)	(0.001)
Mean (at 0.149)	0.785	0.866	35.3	0.145	0.170	0.143
Controls	No	No	No	No	No	No
Observations	146,626	146,626	146,626	146,626	146,626	76,153

2. Research Question and Motivation

Drunk driving remains a major public safety problem in the United States, contributing to more than 585,000 traffic fatalities since 1975. Economic theory suggests that increasing the expected cost of criminal behaviour should reduce offending. In particular, Becker's (1968) model argues that individuals commit crimes when the expected benefits outweigh the expected costs, implying that harsher punishments should deter future criminal activity.

However, estimating the causal effect of punishment severity is challenging because more severe punishments are typically given to more serious offences. Simply comparing offenders who receive harsher and lighter punishments would therefore suffer from omitted variable bias, as differences in future offending may reflect differences in offence severity rather than the punishment itself.

Hansen (2015) overcomes this identification problem by exploiting Washington State's blood alcohol content (BAC) thresholds for driving under the influence (DUI). Drivers with a BAC just below and just above the legal thresholds of 0.08 (DUI) and 0.15 (aggravated DUI) are expected to be very similar, differing primarily in the punishment they receive. This setting allows the author to isolate the causal effect of harsher punishments on future offending. The paper therefore asks whether harsher DUI punishments reduce future drunk driving (recidivism), and more specifically whether crossing the 0.08 BAC threshold reduces future DUI and whether crossing the 0.15 aggravated DUI threshold provides additional deterrence.

3. Data

The analysis uses administrative records from the Washington State Police and Washington State Courts, containing information on 512,964 DUI stops between 1995 and 2011. The dataset includes drivers' blood alcohol content (BAC), demographic characteristics, prior DUI history, court outcomes, punishments, and subsequent DUI offenses.

Although the full dataset spans 1995–2011, the primary analysis focuses on observations from 1999–2007. This period was chosen because Washington adopted fixed legal BAC thresholds on January 1, 1999, with a BAC of 0.08 defining a DUI and 0.15 defining an aggravated DUI. Using data from this period allows the author to examine how crossing either legal threshold affects the likelihood of recidivism within four years of the initial traffic stop.

The sample is restricted to drivers 21 years of age and older, as different BAC limits

apply to underage drivers. BAC is measured to three decimal places, providing the precision needed to compare drivers whose alcohol levels fall just above and just below the legal cutoffs.

The unit of observation is an individual DUI stop. The running variable is the driver's measured BAC, while the treatment variable is a binary indicator equal to one if the driver's BAC exceeds the relevant legal threshold (0.08 or 0.15). Throughout the paper, the primary outcome of interest is whether the driver re-offends within four years. For the Table 2 balance tests replicated here, the dependent variables are predetermined driver characteristics: male, white, age, and accident involvement. The Table 2 specification includes no additional controls; it tests whether these characteristics change discontinuously at the BAC thresholds.

This replication uses a modified version of the original Washington State administrative dataset rather than the confidential data used by Hansen (2015). The modified dataset contains 214,558 observations and includes the key variables required to estimate the regression discontinuity models, such as BAC, DUI indicators, demographic characteristics, accident involvement, and recidivism. However, it does not contain several variables available in the original administrative records, including the driver's prior BAC-test history and portable breath test (PBT) results. Consequently, our replication reproduces only the columns of Table 2 that correspond to the variables available in the modified dataset, resulting in a reduced replication rather than an exact reproduction of the original analysis.

4. Empirical Strategy

The paper estimates the causal effect of harsher DUI punishments using a local linear regression discontinuity (RDD). The key idea is that Washington's DUI laws create two sharp policy thresholds: drivers with a BAC of 0.08 or higher receive a DUI, while those with a BAC of 0.15 or higher receive an aggravated DUI with more severe penalties. Drivers whose BAC falls just below and just above these thresholds are expected to be very similar in both observable and unobservable characteristics, differing primarily in the punishment they receive. Since individuals cannot precisely control or predict their exact BAC before

being tested, assignment around the cutoff is assumed to be essentially random. This allows differences in future recidivism at the thresholds to be interpreted as the causal effect of harsher punishment.

The author estimates the following local linear regression discontinuity model:

$$y_i = X_i'\gamma + \alpha_1 DUI_i + \alpha_2 BAC_i + \alpha_3 (BAC_i \times DUI_i) + u_i \tag{1}$$

where y_i measures whether an individual reoffends, DUI_i is an indicator for being above the relevant BAC threshold, BAC_i is the running variable, and X_i is a vector of control variables including age, gender, race, county, year, and prior offenses. This equation describes the paper's general regression discontinuity framework. For our Table 2 replication, we estimate the no-controls balance-test specification using only the treatment indicator, centered BAC, and their interaction.

The baseline specification uses a local linear regression with a rectangular kernel, while robustness checks are performed using alternative kernels, higher-order polynomials, and different bandwidths. To support the validity of the RDD, the author demonstrates that observable characteristics remain continuous at both the 0.08 and 0.15 BAC thresholds. Within 0.001 BAC of each cutoff, the only statistically significant difference is a small age difference at the 0.15 threshold, providing evidence that drivers on either side of the cutoff are comparable and that the identifying assumptions of the regression discontinuity design are reasonable.

5. Replication Process

We conducted an approximate, reduced replication of Table 2 in Hansen (2015). The goal was to reproduce the regression discontinuity balance tests at the BAC = 0.08 and BAC = 0.15 thresholds using the variables available in the modified course dataset.

We first standardized variable names and converted the variables used in the analysis to

numeric form. We used `low_score` as the BAC running variable. Because this variable was stored in thousandths of BAC, we divided it by 1,000 when necessary so that the running variable was measured in decimal BAC units. We restricted the sample to observations from 1999 through 2007 and to drivers aged 21 or older. We then dropped observations with missing values for BAC, sex, race, age, or accident involvement.

For each threshold, we created a treatment indicator equal to one when BAC was at or above the relevant cutoff. We estimated local linear regressions within a bandwidth of 0.05 BAC units. Each regression included the treatment indicator, BAC centered at the cutoff, and an interaction between the treatment indicator and centered BAC. We clustered standard errors by BAC bins rounded to 0.001.

This is an approximate rather than exact replication because the modified dataset does not contain the *Prior* and *PBT* variables included in the original Table 2. The replication also uses a slightly smaller sample than the published analysis. The Python notebook, input dataset, and output table are available in the accompanying GitHub repository.

6. Replication Results

In this section, we replicate the balance tests from Table 2 in Hansen (2015). These tests do not estimate the effect of harsher punishments on future drunk driving directly. Instead, they test whether drivers just below and just above the BAC thresholds look similar before treatment. This matters because the regression discontinuity design assumes that drivers near each cutoff are comparable, with the main difference being the punishment they receive after crossing the threshold.

We examine the two BAC thresholds used in the paper. At BAC = 0.08, drivers receive a standard DUI. At BAC = 0.15, drivers receive an aggravated DUI, which carries more severe penalties. For each cutoff, we estimate a local linear regression discontinuity model using observations within 0.05 BAC units of the threshold. The regression includes the treatment indicator, BAC centered at the cutoff, and an interaction between the treatment indicator

and centered BAC. We cluster standard errors by BAC values measured to three decimal places, following the original paper.

The original Table 2 includes six predetermined characteristics. Our modified dataset contains four of them: gender, race, age, and whether the stop involved an accident. It does not include the driver's prior BAC-test history or portable breath-test result. For this reason, our results are a reduced replication of Table 2, rather than a full reproduction of every column in the original paper.

Table 1 compares the original estimates with our replication results. The coefficients show whether each predetermined characteristic changes discontinuously when BAC crosses the relevant legal threshold.

Table 1: Comparison of Original and Replicated Regression Discontinuity Balance Tests

	Panel A: DUI Threshold (BAC = 0.08)		Panel B: Aggravated DUI Threshold (BAC = 0.15)	
Characteristic	Original	Replication	Original	Replication
Male	0.007	0.0061	-0.001	-0.0010
	(0.005)	(0.0051)	(0.001)	(0.0040)
White	0.002	0.0017	0.003	0.0047
	(0.005)	(0.0055)	(0.003)	(0.0033)
Age	-0.165	-0.1470	0.049	0.0405
	(0.167)	(0.1658)	(0.123)	(0.1246)
Accident	-0.004	-0.0008	0.003	0.0065
	(0.004)	(0.0035)	(0.004)	(0.0042)
Mean below cutoff: Male	0.792	0.7876	0.785	0.7939
Mean below cutoff: White	0.852	0.8469	0.866	0.8584
Mean below cutoff: Age	34.9	34.9398	35.3	34.1225
Mean below cutoff: Accident	0.089	0.0994	0.145	0.1193
Observations	95,111	93,899	146,626	139,033

Notes: The table compares the published regression discontinuity estimates from Hansen (2015, Table 2) with estimates produced using the modified replication dataset. Each coefficient measures the discontinuity in a predetermined driver characteristic at the relevant BAC threshold. Standard errors clustered by BAC bins of 0.001 are shown in parentheses. Both analyses use a local linear specification with a rectangular kernel and a bandwidth of 0.05 BAC units. The replication includes Male, White, Age, and Accident because the modified dataset does not contain the original paper's Prior BAC Test or Portable Breath Test (PBT) variables.

The replication results are very close to the original results. At the 0.08 DUI threshold, the estimated discontinuities for male, white, age, and accident involvement are all small relative to their standard errors. For example, the coefficient for male is 0.0061 in our replication, compared with 0.007 in Hansen's original table. The estimated age difference is

-0.147 years, compared with -0.165 years in the paper. The estimates for race and accident involvement are also close to the published results.

The same pattern appears at the 0.15 aggravated-DUI threshold. The estimates for male, white, and age are close to the original results, while the estimate for accident involvement is somewhat different in size but remains small and statistically insignificant. Overall, none of the four available characteristics shows a meaningful discontinuity at either BAC threshold.

Our sample is slightly smaller than the one used in the original paper. Around the 0.08 threshold, our replication includes 93,899 observations, compared with 95,111 in the published table. Around the 0.15 threshold, our replication includes 139,033 observations, compared with 146,626 in the original analysis. These differences likely reflect the modified version of the dataset and the available variables. Even with these differences, the coefficients, standard errors, and means below each threshold remain very similar to Hansen's results.

The results support the main identification assumption behind the paper. Drivers just below and just above the BAC thresholds appear similar in terms of gender, race, age, and accident involvement. This gives support to the idea that crossing the BAC threshold changes the punishment a driver receives without separating drivers who were already systematically different before the stop.

7. Conclusion

We replicated the balance tests from Hansen's (2015) study of punishment and deterrence in drunk-driving cases. Using the modified Washington State DUI dataset, we estimated local linear regression discontinuity models around the BAC = 0.08 DUI threshold and the BAC = 0.15 aggravated-DUI threshold. The goal was to test whether predetermined driver characteristics changed at either legal cutoff.

Our results are consistent with the original paper. The estimated discontinuities for male, white, age, and accident involvement are small and statistically insignificant at both thresholds. This suggests that drivers just below and just above the BAC cutoffs are similar

in observable characteristics. Therefore, the results support Hansen's main regression discontinuity assumption that crossing a BAC threshold changes punishment severity without systematically changing the type of driver being compared.

There are limitations to this replication. The modified dataset does not include prior BAC tests or portable breath-test results, so we could not reproduce the full original Table 2. Our sample is also slightly smaller than the original sample. However, the close match between our estimates and the published estimates shows that the available data and Python code reproduce the main balance-test result.

Overall, this replication shows why balance tests are important in regression discontinuity research. Before interpreting a change in recidivism at a BAC threshold as the effect of harsher punishment, we need evidence that drivers on both sides of the cutoff are otherwise similar. Our results provide that evidence for the observable characteristics available in the dataset.

References

[1] Hansen, Benjamin. 2015. "Punishment and Deterrence: Evidence from Drunk Driving." *American Economic Review*, 105(4): 1581–1617. https://doi.org/10.1257/aer.20130189

[2] Becker, G. S. (1968). Crime and Punishment: An Economic Approach. Journal of Political Economy, 76(2), 169–217. https://doi.org/10.1086/259394

[3] Python Software Foundation. 2026. *Python Language Reference*, Version 3.12.

[4] The pandas Development Team. 2026. *pandas Documentation*, Version 2.2.2.

[5] Seabold, Skipper, and Josef Perktold. 2010. "statsmodels: Econometric and Statistical Modeling with Python." *Proceedings of the 9th Python in Science Conference.*

[6] Carrie, Talbot, and Joonas Roose. 2026. "Hansen (2015) Table 2 Replication." GitHub repository. https://github.com/JoonasRoose/hansen-drunk-driving-replication

8. AI Disclosure

ChatGPT was used for coding, debugging, explanations of Python and regression discontinuity methods, and assistance writing and organizing the replication report and repository documentation.

All code, output, and written material were reviewed and checked by Joonas Roose and Talbot Carrie. Both authors understand the final code, data-cleaning steps, regression discontinuity specification, and interpretation of the results.